

Tumors of pharynx

By

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Nasopharyngeal tumors

- more common in Asia, Rare in the US
- Chinese native > Chinese immigrant > North American native
- High index of suspicion required for early diagnosis
- Nasopharyngeal malignancies
 - **SCC (nasopharyngeal carcinoma) 85%**
 - **Lymphoma**
 - **Salivary gland tumors**
 - **Sarcomas**

EPIDEMIOLOGY

Environmental

➤ Viruses

- **EBV- HPV**

➤ Nitrosamines - salted fish

➤ Other recognized risk factors for NPC are cigarette smoking, and occupational exposure to formaldehyde and wood dust.

History:

Cervical LAP 60 %

Nasal obstruction 20%

CHD 20%

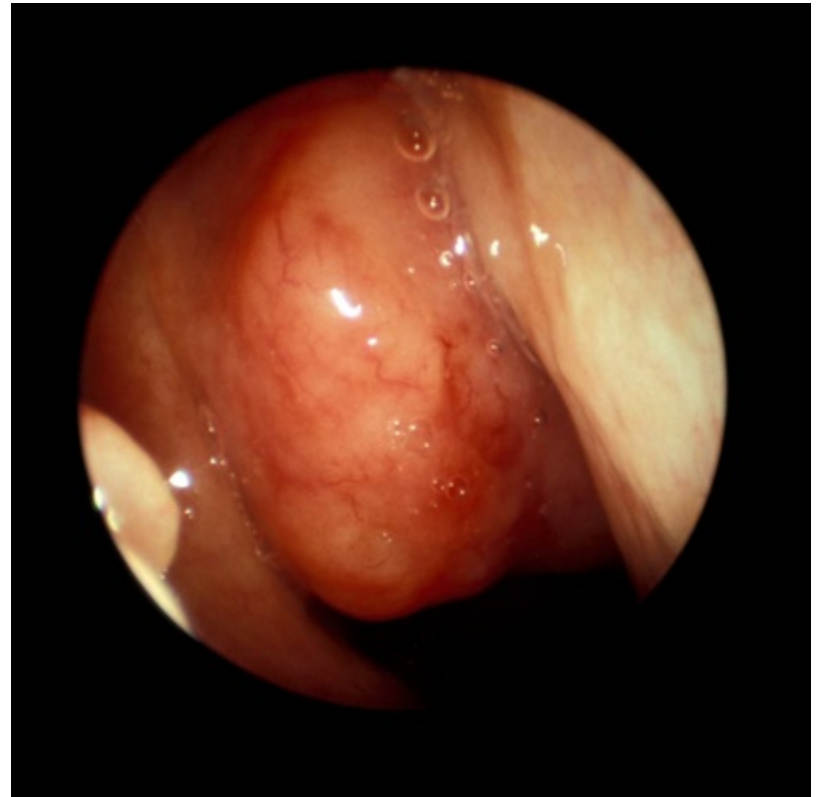
Cranial nerve palsies 10%

Clinical Presentation

- Facial pain - Trigeminal n.
- Diplopia - CN VI
- Ophthalmoplegia - CN III, IV, and VI
- Horner's syndrome - cervical sympathetic
- CN's IX, X, XI, XII - extensive skull base

Examination

- Postnasal mirror
- Endoscopy



- **Nasopharyngeal examination**
 - Fossa of Rosenmuller most common location
 - Variable appearance – exophytic , submucosal
 - NP may appear normal
- **Regional spread**
 - Usually ipsilateral first but bilateral not uncommon
- **Distant spread** - rare (<3%), lungs, liver, bones

Tests for Nasopharyngeal Cancer

- Biopsy
- Endoscopic biopsy
- Fine needle aspiration (FNA) biopsy

Imaging tests

- ❖ Chest x-ray
- ❖ Computed tomography (CT) scan
- ❖ Magnetic resonance imaging (MRI) scan
- ❖ Positron emission tomography (PET) scan

Blood tests

- ✓ Routine blood counts and blood chemistry tests
- ✓ Epstein-Barr virus (EBV) DNA levels

Treatment

- Radiochemotherapy
- Neck dissection

Tumors of Oropharynx

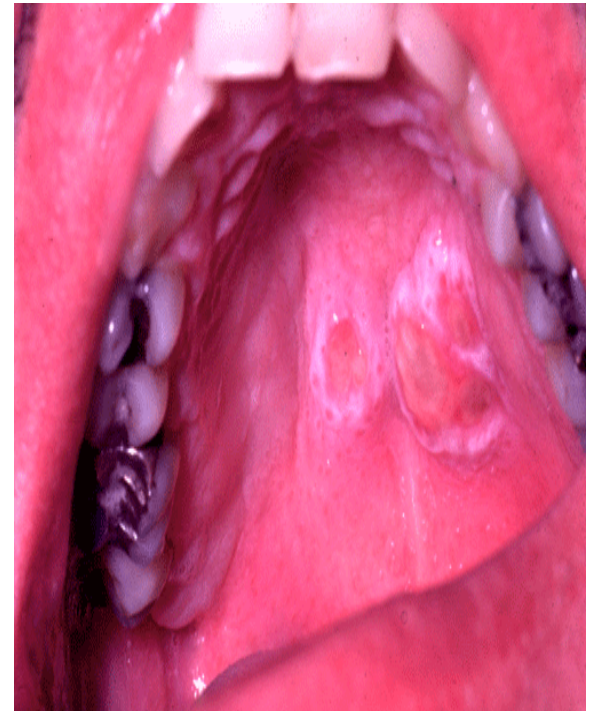
- The oropharynx is lined by squamous epithelium hence squamous cell carcinoma represents the most common tumour.
- However there is abundant lymphoid tissue in the palatine tonsil as well as lingual tonsils, which gets involved with head and neck lymphomas.
- Soft palate is especially rich in minor salivary glands.

Squamous cell carcinoma

- is most common malignancy and forms 90% of tumors of this region.
- The most common sites involved are:
 - Lateral wall (60%)
 - Tongue base (25%)
 - Soft palate (10%)
 - Posterior wall (5%)



- The minor salivary gland tumors have a predilection for soft palate.
- In case of soft palate most minor salivary gland tumours are *pleomorphic adenomas*. Elsewhere malignant tumors are the rule and include *adenoid-cystic* and *muco-epidermoid* types.



Lymphomas

- Lateral wall (tonsils) (90%)
- Tongue base (10%)



Lateral wall tumors

- Most common tumour (60%) and often involves tonsil.
- **Anteriorly** spreads to retromolar trigone, buccal mucosa as well as muscles of tongue base. If the invasion gets deeper the pterygoid muscles are involved resulting in trismus.
- **Lateral spread** involves angle of mandible.
- **Inferiorly** the growth spreads to involve lateral pharyngeal wall and pyriform sinus. The aryepiglottic folds and para-glottic space are involved subsequently.
- The lesions of the lower pole are often difficult to see and some times primary tumours can lurk with in tonsillar crypts as 'occult primaries'

Clinical presentation

- Symptoms frequently do not appear unless lesions are at an advanced stage.

Base of tongue tumors

- 60% to 70% of patients have positive palpable lymph nodes on presentation.
- 20% to 30% have bilateral lymph nodes..
- 20% of patients will present with neck nodes and no apparent primary.

- **The presenting features of oropharyngeal tumours**

- Sore throat
- Otalgia
- Dysphagia
- Ulcers
- Pain
- Trismus
- Neck masses
- Majority of patients present late

Investigations

- US of neck
- CXR
- CT/MRI is done to evaluate tongue base. To see the laterality of the lesion
- It is important to assess any mandibular invasion (Orthopantogram)
- CT scan of chest/abdomen

Biopsy

- Panendoscopy ??? under GA is done to assess size, site and extent of primary tumor, to take a biopsy, to look for metastatic disease and synchronous lesions and to assess neck.
- Incisional biopsy
- If there is smooth regular involvement of tonsil then tonsillectomy
- Deep biopsy for base of tongue

Treatment policy

- **Curative:**

- Radiotherapy
- Surgery
- Surgery plus post-operative radiotherapy

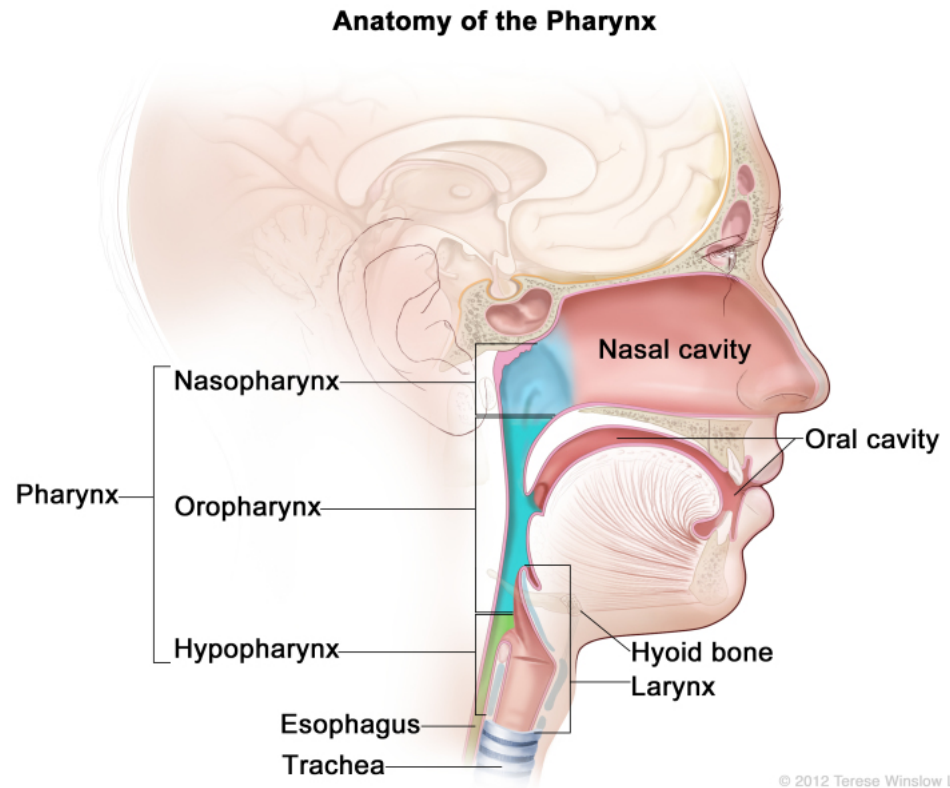
- **Palliative:**

- Radiotherapy
- Radiotherapy and chemotherapy
- Tracheostomy
- Pain relief

Hypopharynx

Anatomical considerations

- It is the lower most part of pharynx.
- It begins at the level of tip of epiglottis and ends at lower border of cricoid cartilage

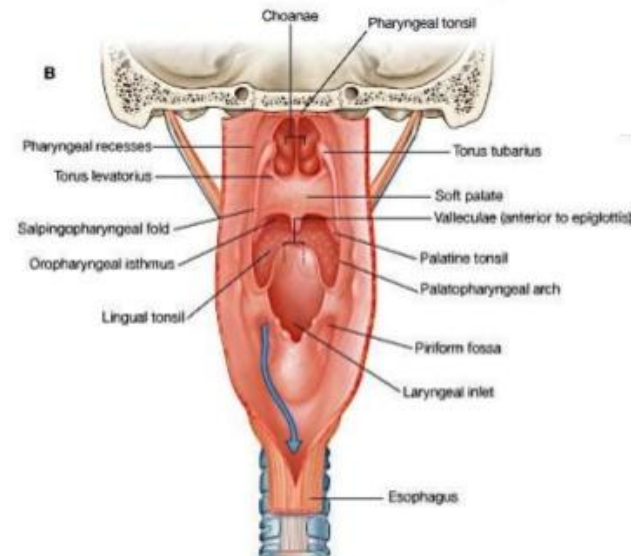


- Tumours arising in this region often present in an advanced state
- Key to cure lies in an early and accurate diagnosis, subsequent staging and in majority cases treatment by curative intent using surgery and post-operative radiotherapy.

- It consists of
 - Two piriform sinuses-one on each side
 - Posterior pharyngeal wall
 - Post-cricoid region where aerodigestive tract continues with cervical oesophagus

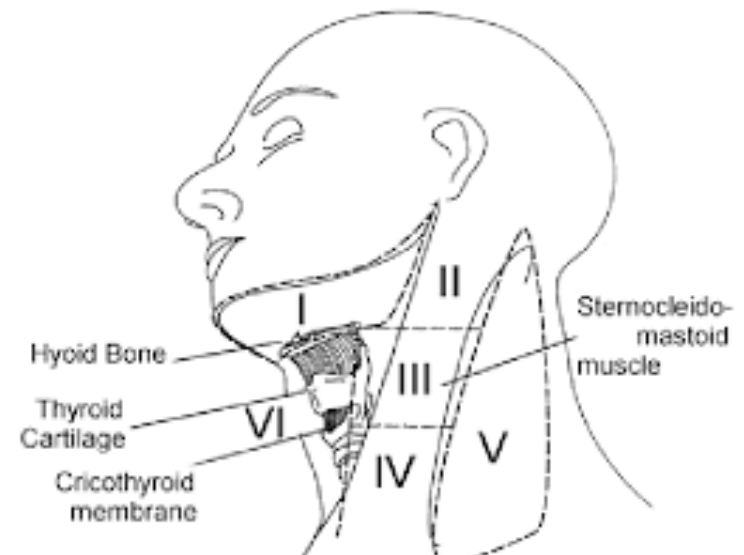
PHARYNX

- Presents openings of
 - auditory tubes
 - two posterior nares
 - larynx
 - Esophagus
- Contains
 - pharyngeal tonsils
 - palatine tonsils
 - lingual tonsils
 - Tubal tonsils



Lymphatics

- Piriform sinus has a rich underlying network of lymphatics but this is less extensive in other subsites.
- In general, the lymphatics in this area drain to deep cervical chain in level IV, but the inferior part of piriform sinus and post-cricoid area also drain to the paratracheal nodes in level VI, and the posterior pharyngeal wall also drains to the retropharyngeal lymph nodes



Surgical Pathology

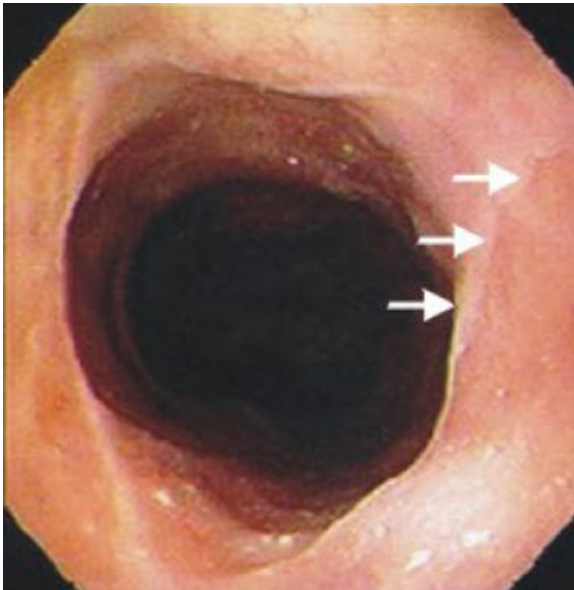
- Almost all carcinomas of the hypopharynx are squamous cell in type.
- Tumours of the piriform sinus constitute the largest group of hypopharyngeal tumours.
- Tumours of the post-cricoid space are the next most common hypopharyngeal.
- Tumours on the posterior hypopharyngeal wall are the least common and constitute approx. 10% of these tumours
- **Alcohol and tobacco** remain the two principal carcinogens implicated in tumours of the upper aerodigestive tract.

Post cricoid carcinoma

- It remains the only squamous cell carcinoma more common in women than men.
- In relation to post cricoid carcinoma a major dietary (iron deficiency) has particularly been described particularly with Plummer Vinson Syndrome.
- This syndrome is also called **Paterson Brown Kelly Syndrome**

Paterson Brown Kelly Syndrome

- Anaemia (both microcytic or macrocytic);
- Glossitis
- Oesophageal web
- Spleenomegaly
- Koilonychia
- Some patients also have history of irradiation of neck either external or internal as radio iodine for thyrotoxicosis.



- **Symptoms suggesting a possible pharyngeal tumours**
- **Dysphagia:** Often persistent and progressive. Patients who complain of food sticking in throat need to be investigated.
- **Pain:** Usually lateralized and prominent on swallowing , may radiate to ipsilateral ear
- **Hoarseness:** When it occurs in association with dysphagia or referred otalgia usually means extension of tumor into larynx

Symptoms suggesting a possible pharyngeal tumour

- **Neck Mass:** Likely to be due to nodal metastasis, but may be due to direct extension through thyrohyoid membrane.
- **Hemoptysis:** An unusual symptom, but can occur with tumors of Piriform Sinus or Post Pharyngeal wall.
- **Weight loss:** Often occurs in presence of significant disease.

Examination

- Indirect Laryngoscopy (IDL)
- Direct Laryngoscopy (DL)

- **Radiological Assessment**

- **Barium Swallow:** Extremely useful investigation in these tumors. Objectives include:
 - ❖ To assess tumor length
 - ❖ To rule out synchronous primary tumor of esophagus
 - ❖ To ascertain presence or absence of aspiration
 - ❖ To assess tumor mobility on vertebral column



CT and MRI

- To assess the extent of the primary tumor and extensions.
- To rule out second primary and distant metastasis
- To assess neck
- To look for cartilage invasion

Endoscopy

- Examination of oral cavity
- Examination of larynx, pharynx, trachea and esophagus
- Biopsy

Treatment policy

- Approximately 25% of patients with hypopharyngeal carcinoma are not treatable at presentation due to advanced age, poor general health, inoperability of the tumor and extensive neck disease. Such patients may be considered for radiotherapy with a palliative intent.

Chemotherapy has no established role.

- Long term results may be obtained with surgery or radiotherapy but are not encouraging.
- 5 year survival rate with surgery and with or w/o post operative radiotherapy are 35%.



Thank
you!!